

Predicting Corporate Bankruptcy from Financial Statements

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Overview

In this project I trained supervised machine learning models that predict from the financial statement indicators of a company whether it will go bankrupt within the next three years, implemented in a Jupyter notebook (modeling.ipynb) with Python, NumPy, Pandas, scikit-learn, Matplotlib and Seaborn. The data is the Polish Companies Bankruptcy Data from the UCI Machine Learning Repository, available at <https://archive.ics.uci.edu/dataset/365/polish+companies+bankruptcy+data> (Tomczak, 2016). A random forest classifier won the comparison against three simpler references, and its scores drive a three-band screening policy that was evaluated once on a held-out test set. The project continues my earlier capstone projects on company register data, and a later project will build a due diligence agent that needs exactly this kind of financial health screening.

Dataset Description

The dataset describes Polish companies from the manufacturing sector through 64 financial indicators, most from one annual statement and a few spanning several years, collected from the Emerging Markets Information Service (Zięba et al., 2016), with bankrupt companies observed from 2000 to 2012 and still operating companies from 2007 to 2013 according to the dataset page (Tomczak, 2016). The archive holds five files, one per forecasting horizon, and I used the largest one, the 3year file, whose label says whether a company went bankrupt within the following three years. This file has 10,503 rows and 65 columns, the 64 anonymous feature

columns Attr1 to Attr64 plus the binary target, and only 495 rows (4.71%) carry the bankrupt label, so the classes are strongly imbalanced. The indicators cover profitability, leverage, liquidity and turnover, for example net profit / total assets (Attr1) and total liabilities / total assets (Attr2). The data checks in the notebook found three quality issues that shaped the preparation: the file is sorted by class, it contains 87 redundant duplicate rows, and 44 of the 64 feature columns have missing values, with more than half of all rows affected.

Modeling Approach

I removed the 87 redundant duplicate rows before any split, because identical rows on both sides of a split would let the model see part of its evaluation data during training. The remaining 10,416 records were split 60/20/20 into training, validation and test sets, stratified by the target and shuffled, because the file is sorted by class. All further preprocessing runs inside the model pipelines of scikit-learn and is fitted on training data only (Pedregosa et al., 2011). Missing values are imputed with the training median, and every affected column gets an additional missing indicator. The data checks suggested that many gaps are structural: a ratio that divides by long-term liabilities has no value for a company without long-term debt, a reading the dataset documentation neither confirms nor contradicts. For a prediction model this is a reasonable choice, because the missing pattern itself can carry information and an important assumption is that the missingness arises the same way in new data, while causal analyses need other methods (Sperrin et al., 2020). The logistic regression additionally standardizes the features, the random forest needs no scaling because tree splits only use the order of the values.

I compared four models of increasing complexity: a dummy baseline that always predicts the majority class, a hand-written solvency rule that flags a company when its net profit is

negative (Attr1 below 0) and more than half of its assets are financed by debt (Attr2 above 0.5), a logistic regression and a random forest with 300 trees. A random forest is an ensemble of many decorrelated decision trees whose combined vote reduces the variance of a single tree (Breiman, 2001). It fits this dataset because the 64 indicators are partly redundant and interact, and because tree splits are not affected by the magnitude of the extreme values in some ratios that may come from near-zero denominators. Both learned models run with balanced class weights, the forest in its per-tree balanced_subsample variant, so the rare bankrupt class carries weight during training. I ran no hyperparameter search. Every configuration was fixed in advance, because the focus of this project is the decision logic and not the last percent of ranking quality.

Accuracy cannot carry the evaluation. A model that never flags anything is right for 95 of 100 companies. The notebook therefore reports precision, recall and F1 for the bankrupt class at the default cut. The selection metric, fixed before any result was computed, is average precision, the mean precision over all thresholds weighted by the gain in recall at each step (scikit-learn developers, 2026). For imbalanced data the precision-recall view is more informative than the ROC curve, because the false positive rate barely reacts to false alarms when the negatives are plentiful (Saito & Rehmsmeier, 2015). Both metrics favor the random forest in my results. I use average precision for selection because the screening task focuses on the rare bankrupt class.

The models bring their own assumptions. The logistic regression assumes a linear relationship between the standardized indicators and the log-odds of bankruptcy (James et al., 2021), and the random forest makes no distributional assumption but, like every supervised model, assumes that future data looks like the training data. The split assumes the rows are exchangeable, which I cannot fully verify because the file carries neither company identifiers nor dates. For the proposed screening use case, I turned the scores of the selected model into a three-

band policy, urgent review, watchlist and no action, with two thresholds chosen on the validation set. The urgent band had to reach a precision of at least 0.50 and both flagged bands together a recall of at least 0.85. Both targets are policy choices, not statistical constants, and in a real deployment they would be set together with the analysts who work the bands. The targets alone do not fix the thresholds, so I used a simple rule on the validation curve. The urgent threshold is the lowest score at which precision reaches 0.50, and the watchlist threshold is the highest score at which recall still reaches 0.85.

Results

The dummy baseline finds no bankruptcies and still reaches 95.30% accuracy on the validation set, which is why accuracy plays no role in the comparison. The solvency rule finds 38 of the 98 validation bankruptcies with 280 false alarms. At the default 0.5 cut the two learned models split the errors in opposite ways, the logistic regression finds 70 of the 98 bankruptcies with 244 false alarms (recall 0.7143, precision 0.2229), while the random forest flags only 15 companies, 12 of them correctly (precision 0.8000, recall 0.1224). Neither is a usable operating point, which is why the selection runs on average precision over all thresholds. The random forest ranks the companies best, with a validation average precision of 0.4687 against 0.3035 for the logistic regression, while ROC-AUC barely separates the two (0.8951 against 0.8721).

Figure 1 shows the forest's precision-recall curve above the logistic regression curve over most of the recall range, and both far above the 4.7% precision of a random ranking. An overfitting check shows that the forest ranks its training data perfectly (training average precision 1.0000). The gap to the validation value is large, which is the usual picture for trees without a

depth limit. Averaging many trees reduces the variance of a single tree (Breiman, 2001), but it does not close this gap. For that reason every model selection and threshold decision in the project uses validation data and never training data.

Figure 1

Precision-Recall Curves of the Two Learned Models on the Validation Set

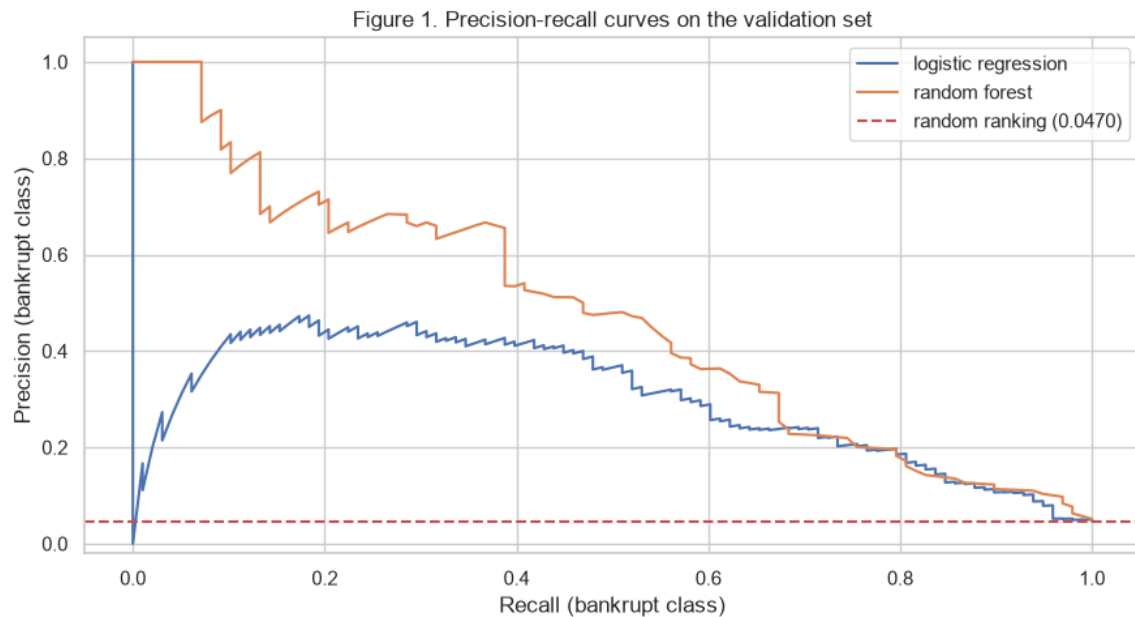
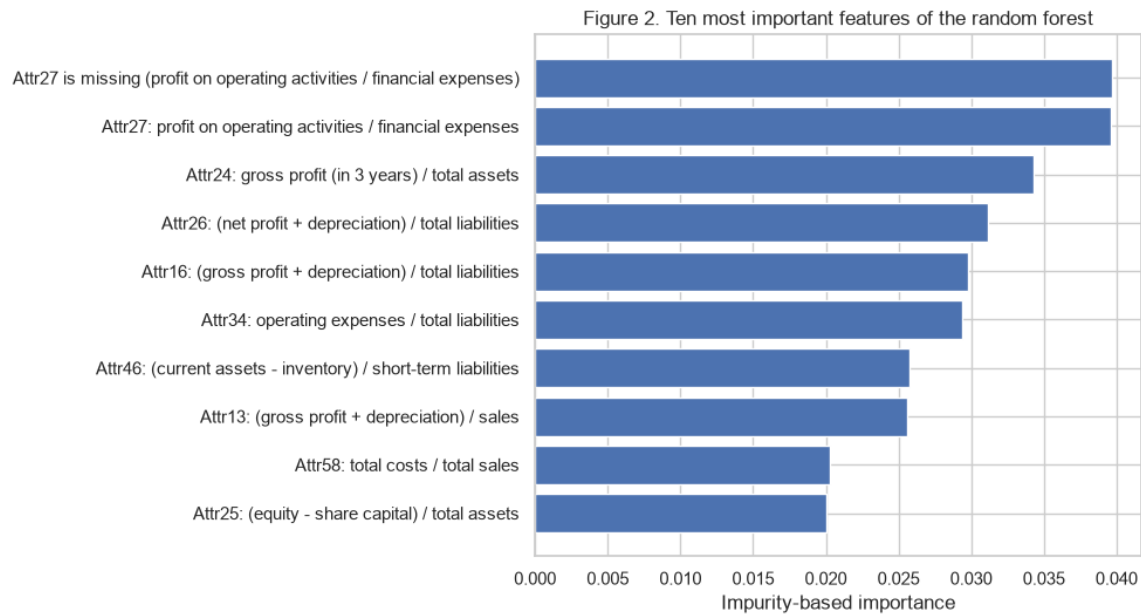


Figure 2 shows the ten features with the highest impurity-based importances in the forest. The list is led by the missing indicator of Attr27 (0.0397), directly followed by Attr27 itself (0.0396). A direct check on the training data shows a strong association, 22.78% of the companies without a value for Attr27 went bankrupt against 3.37% of the companies with a value, while the gap in Attr37, the column with the most missing values, makes no difference (4.75% against 4.73%). Why the value is absent, the source does not say. The ratio divides by financial expenses, so a company without financial expenses is one possible reason, but that remains a possible explanation, not an established finding. No single feature dominates, the top ten of the 108 model inputs, the 64 indicators plus the 44 missing indicators, carry 29.6% of the

total importance. These importances describe the fitted forest on its training data, they do not show causal effects or a stable ranking on unseen data.

Figure 2

Ten Most Important Features of the Random Forest

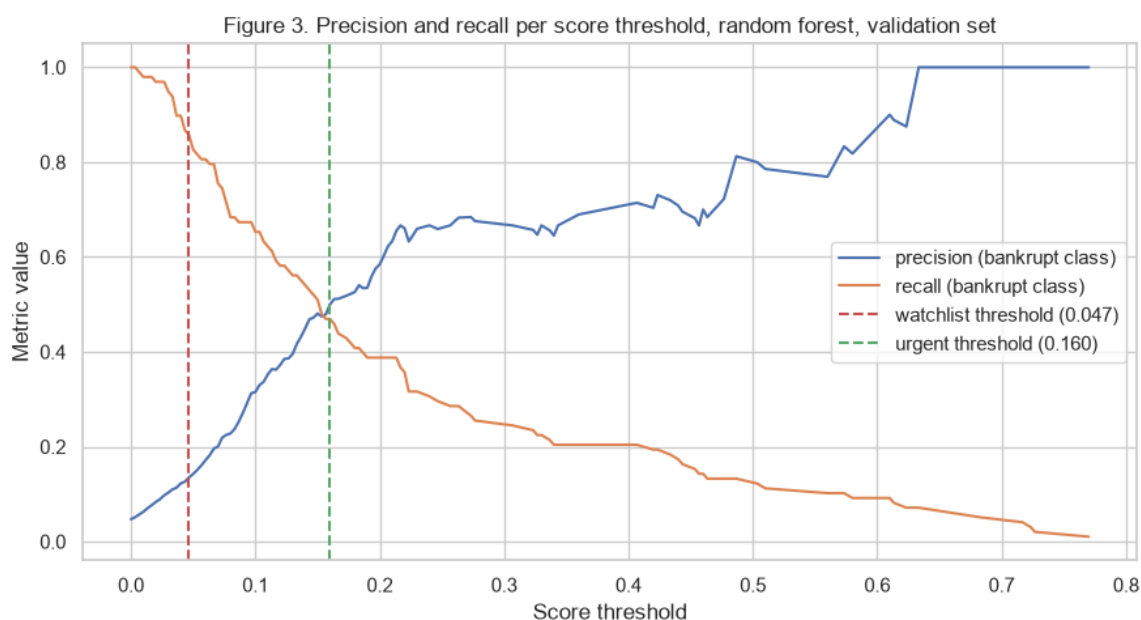


The two policy targets translate into an urgent threshold of 0.160 and a watchlist threshold of approximately 0.04667, shown rounded as 0.047 in Figure 3, which marks where these cuts sit on the precision and recall curves. All band assignments use the unrounded thresholds selected on validation. This matters because 53 validation companies share exactly the watchlist score, and a cut at a rounded value such as 0.047 would drop all of them from the watchlist and miss the recall target. On the untouched test set the frozen policy put 99 of the 2,084 companies into urgent review, 48 of them real future bankruptcies (48.5% against the 50.0% the band reached on validation). The band therefore missed the precision target by 1.5 percentage points. With 99 companies in the band this estimate is uncertain, one company moves the rate by a percentage point, but two more hits would have been needed, so on the held-out

sample the target was not met. The flagged bands together caught 85 of the 99 test bankruptcies (85.9%), so the recall target held on unseen data. The test average precision of 0.5108 and the test ROC-AUC of 0.8917 are close to their validation values (0.4687 and 0.8951). A single split cannot prove more, but it gives no sign that the selection was tuned to a lucky validation set. All numbers come from a fixed random seed of 42, and a full rerun of the notebook with the package versions pinned in requirements.txt reproduces them.

Figure 3

Precision and Recall per Score Threshold With the Two Chosen Cuts on the Validation Set



Interpretation for a Non-Technical Audience

The model reads 64 numbers from a company's financial statements and turns them into a risk score. In the held-out test sample, about 5 of every 100 company records were assigned to the urgent-review band, and about every second company in this small group really went bankrupt within the next three years. Another 25 landed on a watchlist for the next periodic review, and the remaining 70 passed without action, with fewer than 1 in 100 of them failing

later. In total the screening caught about 86 of every 100 companies that actually failed, at the cost of many watchlist entries that turned out fine. These shares describe one historical sample of Polish manufacturing companies. They are not guaranteed hit rates for new companies, because the rates also depend on how many failing companies are in the group being screened.

The score is a warning system, not a verdict. A high score means that the company's numbers look like the numbers of companies that failed before. It does not prove that this company will fail, and the final judgment stays with a human analyst who sees more than 64 numbers.

Limitations and Potential Bias

The clearest limitation is the scope of the data. The companies are Polish manufacturing companies observed between 2000 and 2013, so the model has learned the bankruptcy patterns of one sector, one economy and one period, and any use on other sectors such as services, on another market or on a later period needs a re-validation on local data. The two observation periods are also not settled. The dataset page assigns 2000 to 2012 to the bankrupt companies and 2007 to 2013 to the operating ones (Tomczak, 2016), while Zięba et al. (2016) state them the other way round, and the file carries no dates to check this. A second limitation sits in the file itself. It carries no company identifiers and no dates, so I cannot verify whether related records cluster inside my split, and the extreme values in several ratios that may come from near-zero denominators cannot be checked against the underlying statements. A third limitation is the width of the watchlist, catching 85% of the bankruptcies forces a quarter of all companies onto it, and its bankruptcy rate of 7.2% on validation is only modestly above the 4.7% base rate.

Bias usually enters a model through the data it learns from (Barocas & Selbst, 2016), and here the main risk is a selection and label bias in the source. The dataset only contains companies

with usable financial statements, and the source does not describe how the bankrupt label was built in enough detail to tell whether companies that failed informally or simply stopped filing are captured. Such companies would be invisible to the model. A second risk appears in deployment, an analyst who trusts the score too much would investigate exactly the companies the model flags, which makes the model look more right than it is. As a concrete mitigation step the screening keeps a human in the loop by design, the three bands only prioritize human review and no final adverse decision about a company is made automatically. To check for missed cases and for selective follow-up, I would also review a random sample from the no-action band and collect the outcome labels independently of the flags of the model. Before any real use I would re-validate the model and both thresholds on freshly labeled local data at fixed intervals.

A useful next step would be a gradient boosting model, for which Zięba et al. (2016) report the best results on this dataset, together with a calibration of the scores, after which both thresholds would be chosen again on the calibrated probabilities. The other four horizon files of the archive follow the same forecasting design in other years with a different label. Because the files carry no company identifiers, an overlap of companies between them cannot be ruled out, so they are no independent test sets without further information from the source.

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